

William Marella

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Senior data scientist specializing in causal inference, statistical measurement, experimentation, and applied AI. Led analytical strategy for a \$10M ARPA-H program and invented a patent-pending epigenetic aging algorithm commercially licensed to TruDiagnostic.

EDUCATION

University of Cambridge

November 2024

M. Phil. Population Health Sciences, Specialization in Health Data Science

Brown University

May 2023

B.A. Biology, GPA: 3.96

WORK EXPERIENCE

Senior Data Scientist

New York, New York

Columbia University, Geroscience Computational Core

September 2024 - Present

- Lead analytics and engineering for the Core, translating ambiguous scientific questions and collaborator needs into causal-inference pipelines, ML workflows, statistical software, and decision-ready analyses for technical and nontechnical stakeholders.

ARPA-H FAST Grant | *R, Python, Causal Inference, ML*

- Led design and execution of causal inference and ML strategy for a **\$10M ARPA-H grant**; architected collaborator-facing R/Python pipelines used by clinical and industry partners including **Novo Nordisk and Lilly**.
- Translated grant aims and scientific questions into reproducible, portable analytical workflows: treatment-effect estimation and mediation analysis for high-dimensional clinical and omics data, and ML-based prediction of clinical endpoints from repeated-measures omics.

FraminghamPACE | *R, Hierarchical Modeling, ML* | [medRxiv](#)

- Developed a hierarchical model of the pace of aging and machine learning training methods to create a **state-of-the-art epigenetic aging clock** with improved predictive performance, technical reliability, and responsiveness to intervention.
- First author** of the resulting manuscript, currently in review; **inventor** of the patent-pending algorithm, **commercially licensed** by Columbia University to TruDiagnostic.

QueryGaP | *Python, PostgreSQL, TypeScript, Next.js* | [Website](#)

- Built and deployed an **LLM-powered search platform** for dbGaP and UK Biobank documentation, reducing phenotype-variable discovery from hours to minutes and gaining adoption at Columbia and external institutions.
- Scraped, normalized, and indexed large-scale study and variable documentation in PostgreSQL, combining semantic retrieval and full-text search with pgvector and tsvector.

Visiting Researcher

Cambridge, UK

University of Cambridge, Inouye Lab

June 2024 - June 2026

- Derived mean-field variational inference and partially collapsed Gibbs sampling algorithms for Latent Factor Allocation, an unsupervised Bayesian machine learning method; benchmarked their accuracy and computational complexity on simulated data designed to resemble UK Biobank data.
- Implemented both algorithms from scratch in NumPy/SciPy, including convergence diagnostics and multithreaded MCMC chain coordination, and released the work as a [public Python library](#).

SELECTED PRESENTATIONS

- NIA Biomarker Network Meeting 2026** | St. Louis, MO | Invited Oral Presentation
FraminghamPACE: A New DNA Methylation Biomarker of the Pace of Aging
- Biomarkers of Aging Consortium 2025** | Boston, MA | Poster Presentation
FraminghamPACE: An Epigenetic Speedometer of the Pace of Aging

SELECTED AI & SOFTWARE PROJECTS

LLM Agents Play Nash | *Python, OpenAI SDK* | [Technical Writeup](#), [GitHub](#)

- Built a **multi-agent simulation framework** demonstrating that frontier LLM agents can converge to Nash equilibria in novel, non-canonical repeated strategic environments, with performance strongly gated by inference-time reasoning budget.

- Designed repeated-game environments with partial information and explicit payoff feedback, structuring prompts and context to support strategic adaptation across rounds.
- Instrumented the framework to log beliefs, actions, payoffs, reasoning traces, latency, and inference cost, enabling diagnosis of failure modes in payoff calculation, best-response reasoning, and equilibrium convergence.

OpenCode Protect | *TypeScript, Node.js, Bun, Unix/Linux* | [GitHub](#)

- Forked OpenCode to implement **OS-level isolation for AI agent-generated shell commands**, running tool execution under a restricted Unix user to protect credentials and sensitive files.
- Used Unix permissions as a security boundary independent of model behavior, constraining agent access even when malicious repository content or tool outputs bypass prompt-level safeguards.

ASCII Doodle | *Python, PyTorch, Transformers* | [GitHub](#), [Hugging Face](#)

- Text-to-ASCII art generator powered by a **300M-parameter Diffusion Transformer (DiT)** trained with flow matching over a **VAE-compressed latent space** of 32x16 ASCII token grids.
- Trained on Quick, Draw! using NVIDIA A100 GPUs on Modal and published the model weights on Hugging Face.
- Conditioned the model on **CLIP text embeddings** with classifier-free guidance to control prompt adherence; demonstrated generalization to synonyms and near-out-of-distribution prompts unseen during training.

CostcoGuessr | *TypeScript, Next.js, Bun, PostgreSQL* | [Website](#)

- Built and **deployed a full-stack daily price-guessing game** with a TypeScript/Bun backend, Next.js frontend, PostgreSQL database, and image storage on Vercel Blob.
- **Scraped and normalized 10,000+ Costco products across 11 countries**, including product metadata, prices, and images.
- Grew organically to **200 daily users**.

Open-Source Contributions: *TypeScript, React, Bun, Node.js* | [OpenCode PR](#), [PostHog PR](#)

- Merged navigation and UI features and fixes into **OpenCode** and **PostHog** codebases

SKILLS

Languages & Systems: Python, R, SQL, TypeScript, JavaScript, Bash, Unix, Linux

Experimentation & Causal Inference: experimentation, A/B testing, metric design, treatment-effect estimation, mediation analysis, quasi-experimental design, difference-in-differences, propensity score methods, sensitivity analysis, hypothesis testing, power analysis

Statistical Modeling: mixed-effects models, survival analysis, longitudinal modeling, Bayesian inference, hierarchical models, MCMC, variational inference, multiple testing, false discovery rate (FDR), model diagnostics

Machine Learning: NumPy, SciPy, PyTorch, Hugging Face Transformers, scikit-learn, supervised and unsupervised learning, regularized regression, gradient boosting, Optuna, feature selection, model interpretability, NLP, text mining, clustering, dimensionality reduction

AI Systems & Evaluation: LLM applications, retrieval-augmented generation (RAG), semantic search, vector search, LangChain, LangGraph, AI agents, agentic workflows, prompt engineering, context engineering, LLM evaluation, model behavior analysis, synthetic data generation, multi-agent simulation

Data & Software Engineering: PostgreSQL, pgvector, APIs, data pipelines, web scraping, Docker, Git, GitHub, reproducible analysis, Next.js, React, Node.js, Bun, Vercel, Railway